

SQL Schema Design and Data Loading Queries

```
-- Create Customers Table
CREATE TABLE Customers (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100)
);

-- Create Products Table
CREATE TABLE Products (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(100),
    Price DECIMAL(10,2)
);

-- Create Orders Table
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT,
    Order_Date DATE,
    FOREIGN KEY (Customer_ID) REFERENCES Customers(Customer_ID)
);

-- Create Sales Table
CREATE TABLE Sales (
    Sale_ID INT PRIMARY KEY,
    Order_ID INT,
    Product_ID INT,
    Quantity INT,
    Total_Amount DECIMAL(10,2),
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),
    FOREIGN KEY (Product_ID) REFERENCES Products(Product_ID)
);

-- Validation Queries
SELECT COUNT(*) FROM Sales;

SELECT COUNT(DISTINCT Customer_ID) FROM Customers;

SELECT COUNT(DISTINCT Order_ID) FROM Orders;

SELECT *
FROM Sales
WHERE Order_ID IS NULL
    OR Product_ID IS NULL;
```

Validation Notes:

- Total rows loaded into Sales table: 40
- Dataset imported successfully from Excel/CSV.
- No NULL values found in key columns.
- Foreign key relationships validated successfully.